

RISHAB KUMAR

Nalanda, Bihar 803110

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Education

Indian Institute Of Technology Madras

Bachelor of Science in Data Science and Its Applications

Expected May 2027

Chennai, Tamil Nadu

Patliputra University

Bachelor of Arts (Sociology)

Completed(2023)

Patna, Bihar

Projects

Predicting Irrigation Need | *Python, Scikit-Learn, LightGBM, SHAP, Pandas, NumPy*

- Engineered a precision agriculture machine learning solution utilizing Scikit-Learn and Pandas to predict optimal irrigation schedules across 638304 data points.
- Trained and fine-tuned a LightGBM classification model, implementing cross-validation and advanced imbalance handling to achieve Average Balanced Accuracy: 0.9607 ± 0.0016 .
- Integrated Explainable AI (SHAP) to interpret complex model predictions, extracting actionable insights regarding key environmental factors like Crop Growth Stage, Soil Moisture, Mulching used and wind speed.
- Established modular data preprocessing pipelines and standardized workflows in Google Colab, ensuring end-to-end project reproducibility.
- Github Repository:** <https://github.com/RishabKrPandey/Predicting-Irrigation-Needs>

Kaggle Competition: Engage2 Value | *Python, NumPy, Scikit-Learn, LightGBM, Matplotlib, Seaborn*

- Engineered a predictive model utilizing LightGBM with Tweedie loss to accurately forecast customer purchase value from raw click data.
- Secured a competitive global ranking of 179th out of 1,790 participants on the Kaggle leaderboard by achieving an accuracy score of 0.677.
- Conducted extensive exploratory data analysis (EDA) using Matplotlib and Seaborn to identify user behavioral patterns and engineer high-value predictive features.
- Github Repository:** <https://github.com/RishabKrPandey/Kaggle-Notebook-Engage2Value>

Chocolate-Sales-Analysis | *Python, Pandas, Matplotlib, Seaborn, Google Colab*

- Spearheaded comprehensive Exploratory Data Analysis (EDA) on a chocolate sales dataset using Python, cleaning raw financial data to evaluate product performance across 6 global regions.
- Engineered data visualizations to uncover critical business patterns, successfully identifying that the top 3 products alone generated over \$3.2 Million in revenue and that peak seasonality occurred around January.
- Translated complex data trends into strategic business intelligence, delivering recommendations to increase marketing spend in high-demand regions like the UK and India while optimizing inventory to prevent stockouts of top-tier items.
- Github Repository:** <https://github.com/RishabKrPandey/Chocolate-Sales-Analysis>

Household Service Application | *Python, Flask, Vue.js, SQLite, Redis, Celery, Chart.js*

- Architected a multi-user, full-stack web application utilizing a Flask RESTful API backend and a responsive Vue.js frontend to manage household service requests.
- Engineered an asynchronous task queue using Celery and Redis to reliably process background operations such as daily automated email notifications or batch report generation, significantly improving main-thread response times.
- Implemented strict Role-Based Access Control (RBAC) via Flask-Security, ensuring secure authentication and localized data privacy for admins, service professionals, and standard users.
- Designed an interactive analytics dashboard integrating Chart.js to visualize key user statistics, enabling administrators to track service completion and other details
- Github Repository:** <https://github.com/RishabKrPandey/Household-Services-App>

Technical Skills

Languages: Python, SQL, JavaScript, HTML/CSS

Data Science & Machine Learning: Pandas, NumPy, Scikit-Learn, LightGBM, Random Forest, XGBoost, Gradient Boosting, Matplotlib, Seaborn, SHAP

Web Frameworks & Technologies: Flask, Flask-RESTful, Vue.js, Redis, Celery

Tools & Databases: Git/GitHub, SQLite, Jupyter Notebook, VS Code

Leadership / Extracurricular

Secretary, Quiz Society (2023–24) — Interests: Volleyball, Javelin